C. F. Gauss / R. Dedekind

De nexu inter multitudinem classium

English translation of Dedekind's remarks, printed pp. 292–303

Remarks on the memoir *De nexu inter multitudinem classium, in quas formae binariae secundi gradus distribuuntur, earumque determinantem*

On I and II

The second formula for the number of points lying inside the circle is obtained by considering the square inscribed in the circle with its sides parallel to the coordinate axes. Comparing the two formulae leads to the arithmetically elementary identity

$$r' + r'' + \cdots + r^{(q)} = q^2 + r^{(q+1)} + r^{(q+2)} + \cdots + r^{(r)}.$$

From it one also obtains the first of the two following rules found on a separate sheet:

$$1 + 4\sqrt{A} + 4\sqrt{\frac{1}{4}A} + 8\sum\left(\sqrt{A - n^2} - n\right),$$

where in each square root the fractional part is to be omitted and the summation is from $n = 1$ to $n = \sqrt{A}$, that is, to $\sqrt{A/4}$. The other formula is

$$1 + 4\left\{A - \frac{A}{3} + \frac{A}{5} - \frac{A}{7} + \frac{A}{9} - \frac{A}{11} + \cdots\right\},$$

again with the fractional part omitted in each term. This latter formula follows from the later theorem on the number of representations of a given number by the form $x^2 + y^2$. That theorem may be stated as follows: the number of different representations of a positive integer m by $x^2 + y^2$ is $4(a - b)$, where a, b count the divisors of m of the forms $4n + 1$ and $4n + 3$, respectively. Comparing this arithmetic formula with the mean number of representations obtained geometrically gives rigorously

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots.$$

On III and IV

Let C be the complex of all positive properly primitive forms of negative determinant $-D$, with no two properly equivalent. If the variables of these forms are given pairs of relatively prime integer values, the number of representations of a positive integer m is $\varepsilon\psi(m)$. Here ε is the number of solutions of

$$t^2 + Du^2 = 1,$$

and $\psi(m)$ is the number of roots n of

$$n^2 + D \equiv 0 \pmod{m}$$

for which m, $2n$, and $(n^2 + D)/m$ have no common divisor. The factor ε is 4 for $D = 1$ and 2 otherwise. If $m = p^{\pi} p'^{\pi'} p''^{\pi''} \cdots$, then

$$\psi(m) = \psi(p^{\pi})\psi(p'^{\pi'})\psi(p''^{\pi''}) \cdots .$$

If $\mathfrak{A}(m)$ denotes the number of roots of $n^2 + D \equiv 0 \pmod{m}$, then

$$\psi(p^{\pi}) = \mathfrak{A}(p^{\pi}) = 1 + \left(\frac{-D}{p}\right)$$

when p does not divide $2D$, while otherwise

$$\psi(p^{\pi}) = \mathfrak{A}(p^{\pi}) - \frac{1}{p}\,\mathfrak{A}(p^{\pi+1}).$$

If all integer values are admitted for the variables of the forms in C, the number of representations is $\varepsilon f(m)$, where

$$f(m) = \sum \psi\left(\frac{m}{\mu^2}\right),$$

the sum extending over all square divisors μ^2 of m. Thus

$$f(m) = f(p^{\pi})f(p'^{\pi'})f(p''^{\pi''}) \cdots$$

and

$$f(p^{\pi}) = \psi(p^{\pi}) + \psi(p^{\pi-2}) + \psi(p^{\pi-4}) + \cdots .$$

If $p \nmid 2D$, this gives

$$f(p^{\pi}) = 1 + \left(\frac{-D}{p}\right) + \left(\frac{-D}{p^2}\right) + \cdots + \left(\frac{-D}{p^{\pi}}\right),$$

and, when m is relatively prime to $2D$,

$$f(m) = \sum_{n|m} \left(\frac{-D}{n}\right).$$

These observations justify the statements in the text about the number (m), with the additional restriction in the first statement that D not be divisible by p^2. They also give the value of

$$1 + \frac{f(p)}{p} + \frac{f(p^2)}{p^2} + \frac{f(p^3)}{p^3} + \cdots$$

as

$$\frac{1}{1 - \frac{1}{p}} \quad \text{or} \quad \frac{1}{1 - \frac{1}{p}} \cdot \frac{1}{1 - \left(\frac{-D}{p}\right)\frac{1}{p}},$$

according as p divides $2D$ or not.

On V

The note attached to Formula III indicates the route by which Gauss arrived at the number k of forms contained in the complex C. Geometrically, the limiting mean number of representations is

$$k\frac{\pi}{\sqrt{D}}.$$

An arithmetical expression for the same mean is obtained from $(m) = \varepsilon f(m)$. Dedekind describes Gauss's apparent procedure: successively eliminate the primes from $f(m)$. If $\theta'(m)$ is obtained from $\theta(m)$ by eliminating a prime p, then, when $p \nmid 2D$,

$$\theta'(m) = \theta(mp) - \left(\frac{-D}{p}\right)\theta(m),$$

and if ω, ω' are the mean values of θ, θ', then

$$\omega = \frac{\omega'}{1 - \left(\frac{-D}{p}\right)\frac{1}{p}}.$$

When $p \mid 2D$, the mean values are equal. Hence the mean value of $f(m)$ is formally

$$\prod_p \frac{1}{1 - \left(\frac{-D}{p}\right)\frac{1}{p}},$$

where p runs through the primes not dividing $2D$, and consequently

$$k = \frac{\varepsilon\sqrt{D}}{\pi}\prod_p \frac{1}{1 - \left(\frac{-D}{p}\right)\frac{1}{p}}.$$

Dedekind then points out a gap in rigor: the convergence of the mean values after infinitely many eliminations is not automatic. He explains a direct way around this. Let

$$\theta(m) = \sum \left(\frac{-D}{n}\right),$$

where n runs through the divisors of m relatively prime to $2D$. Then

$$\Theta(m) = \theta(1) + \cdots + \theta(m) = \sum \left(\frac{-D}{n}\right)\left\lfloor \frac{m}{n} \right\rfloor.$$

Ordered by increasing n, the remaining tail after the term $\left(\frac{-D}{n}\right) m/n$ is bounded by $\varphi(2D)m/n$, because sums of complete periods of the character vanish. Thus

$$\sum \left(\frac{-D}{n}\right)\frac{1}{n}$$

is the limiting mean value, and therefore

$$k = \frac{\varepsilon\sqrt{D}}{\pi}\sum \left(\frac{-D}{n}\right)\frac{1}{n}.$$

On VI and VII

For $D \equiv 3 \pmod 4$, using

$$\operatorname{cotang} u = \frac{1}{u} + \frac{1}{u-\pi} + \frac{1}{u+\pi} + \frac{1}{u-2\pi} + \frac{1}{u+2\pi} + \cdots$$

one obtains

$$\sum \left(\frac{-D}{n}\right)\frac{1}{n} = \frac{\pi}{2D}\sum_{\nu}{}' \left(\frac{\nu}{D}\right) \operatorname{cotang} \frac{\nu\pi}{2D}.$$

Putting $\sqrt{-1} = i$ and $r = \cos(2\pi/D) + i\sin(2\pi/D)$, this becomes

$$\sum \left(\frac{-D}{n}\right)\frac{1}{n} = \frac{\pi i}{4D}\left(\frac{2}{D}\right)\sum \left(\frac{\mu}{D}\right)\frac{r^{\mu}-1}{r^{\mu}+1}.$$

With

$$\frac{\omega-1}{\omega+1} = \sum (-1)^{\alpha-1}\omega^{\alpha}$$

and Gauss's summation theorem, one obtains, for square-free D,

$$\sum \left(\frac{-D}{n}\right)\frac{1}{n} = \frac{\pi}{2\sqrt{D}}\sum \left(\frac{\alpha'}{D}\right),$$

where α' runs through the reduced residues below $D/2$. Since $\varepsilon = 2$, this gives

$$k = \sum \left(\frac{\alpha'}{D}\right).$$

For $D \equiv 1 \pmod 4$, the corresponding use of the cosecant expansion gives

$$\sum \left(\frac{-D}{n}\right)\frac{1}{n} = \frac{\pi}{2D}\sum \left(\frac{\mu}{D}\right)\frac{r^{\mu}}{r^{2\mu}+1}.$$

Using

$$\frac{\omega}{\omega^2+1} = 1 + \sum \omega^{4\alpha''} + \sum \omega^{D-4\alpha''},$$

one obtains

$$\sum \left(\frac{-D}{n}\right)\frac{1}{n} = \frac{\pi}{\sqrt{D}}\sum \left(\frac{\alpha''}{D}\right)$$

and therefore

$$k = 2\sum \left(\frac{\alpha''}{D}\right).$$

Dedekind adds that the cases with even D can be treated similarly. For positive determinants D, the extant manuscript material contained only hyperbolic-sector figures and inequalities defining the admissible representations. These inequalities isolate one representation among infinitely many equivalent ones; the resulting representation count is given by the same function $f(m)$, with $-D$ replaced by D.

On VIII

Here p is a positive prime of the form $4n+1$, and the notation agrees with that of *Theoria residuorum biquadraticorum*, first memoir, Article 23:

$$f \equiv 1 \cdot 2 \cdot 3 \cdots \frac{1}{2}(p-3) \cdot \frac{1}{2}(p-1) \pmod{p},$$

$$p = a^2 + b^2, \qquad a \equiv 1 \pmod{4}, \qquad b \equiv af \pmod{p}.$$

The column f was added to the two tables found among the papers, and some gaps in them were filled.

The theorem in the text is connected with the biquadratic character of 2. Since

$$2^{\frac{p-1}{4}} \equiv f^{b/2} \pmod{p},$$

the congruence $b \equiv 2m + a - 1 \pmod{8}$ is equivalent to

$$2^{\frac{p-1}{4}} \equiv f^{m+\frac{a-1}{2}} \pmod{p}.$$

Dedekind gives separate proofs for $p \equiv 1 \pmod{8}$ and $p \equiv 5 \pmod{8}$, using products A and B of the corresponding quadratic residues and non-residues, and obtains in both cases the same congruence.

On IX

Let p be positive, odd, and square-free, and let

$$S_r = \sum \left(\frac{s_r}{p}\right),$$

where s_r runs through the reduced residues lying between $(r-1)p/8$ and $rp/8$. If C_1 and C_2 denote the numbers of primitive proper classes for determinants $-p$ and $-2p$, respectively, then

$$C_1 = 2(S_1 + S_2), \qquad C_2 = 2(S_1 - S_4),$$

or

$$C_1 = S_1 + S_2 + S_3 + S_4, \qquad C_2 = 2(S_2 + S_3),$$

according as $p \equiv 1$ or $3 \pmod{4}$. From the doubling relations between the intervals one obtains, when $p \equiv 1 \pmod{4}$,

$$\begin{aligned}
S_1 = S_8 &= \frac{1}{4}\left(\frac{2}{p}\right) C_1 + \frac{1}{4} C_2, \\
S_2 = S_7 &= \frac{1}{4}\left(2 - \left(\frac{2}{p}\right)\right) C_1 - \frac{1}{4} C_2, \\
S_3 = S_6 &= -\frac{1}{4}\left(2 + \left(\frac{2}{p}\right)\right) C_1 + \frac{1}{4} C_2, \\
S_4 = S_5 &= \frac{1}{4}\left(\frac{2}{p}\right) C_1 - \frac{1}{4} C_2,
\end{aligned}$$

and, when $p \equiv 3 \pmod 4$,

$$\begin{aligned}
S_1 = -S_8 &= \frac{1}{4}\left(3 + \left(\frac{2}{p}\right)\right) C_1 - \frac{1}{4}C_2,\\
S_2 = -S_7 &= -\frac{1}{4}\left(1 - \left(\frac{2}{p}\right)\right) C_1 + \frac{1}{4}C_2,\\
S_3 = -S_6 &= \frac{1}{4}\left(1 - \left(\frac{2}{p}\right)\right) C_1 + \frac{1}{4}C_2,\\
S_4 = -S_5 &= \frac{1}{4}\left(1 - \left(\frac{2}{p}\right)\right) C_1 - \frac{1}{4}C_2.
\end{aligned}$$

For prime p this immediately yields the formulas printed in the preceding text for the number of quadratic residues in each octant.

On X

For twelfths, let p be square-free of the form $6n \pm 1$ and define S_r analogously with intervals of length $p/12$. If C_1, C_3 correspond to determinants $-p, -3p$, then

$$C_1 = 2(S_1 + S_2 + S_3), \qquad C_3 = 2(S_1 + S_2 - S_5 - S_6),$$

or

$$C_1 = S_1 + \cdots + S_6, \qquad C_3 = 2(S_2 + S_3 + S_4 + S_5),$$

according as $p \equiv 1$ or $3 \pmod 4$. Combining the relations from multiplication by 2 and 3 gives, for $p \equiv 1 \pmod 4$,

$$\begin{aligned}
S_1 = S_{12} &= \frac{1}{4}\left(1 + \left(\frac{3}{p}\right)\right) C_1 + \frac{1}{12}\left(1 + \left(\frac{2}{p}\right)\right) C_3,\\
S_2 = S_{11} &= -\frac{1}{4}\left(1 + \left(\frac{3}{p}\right)\right) C_1 + \frac{1}{12}\left(1 + \left(\frac{2}{p}\right)\right) C_3,\\
S_3 = S_{10} &= \frac{1}{2}C_1 - \frac{1}{6}\left(1 + \left(\frac{2}{p}\right)\right) C_3,\\
S_4 = S_9 &= -\frac{1}{2}C_1 - \frac{1}{6}\left(2 - \left(\frac{2}{p}\right)\right) C_3,\\
S_5 = S_8 &= \frac{1}{4}\left(1 + \left(\frac{3}{p}\right)\right) C_1 - \frac{1}{12}\left(5 - \left(\frac{2}{p}\right)\right) C_3,\\
S_6 = S_7 &= -\frac{1}{4}\left(1 + \left(\frac{3}{p}\right)\right) C_1 + \frac{1}{12}\left(1 + \left(\frac{2}{p}\right)\right) C_3,
\end{aligned}$$

and for $p \equiv 3 \pmod 4$,

$$
\begin{aligned}
S_1 = -S_{12} &= \frac{1}{12}\left(9 + 3\left(\frac{2}{p}\right) - \left(\frac{3}{p}\right) + \left(\frac{6}{p}\right)\right) C_1 - \frac{1}{4}C_3,\\
S_2 = -S_{11} &= \frac{1}{4}\left(-1 + \left(\frac{2}{p}\right) + \left(\frac{3}{p}\right) - \left(\frac{6}{p}\right)\right) C_1 + \frac{1}{4}C_3,\\
S_3 = -S_{10} &= \frac{1}{6}\left(-\left(\frac{3}{p}\right) + \left(\frac{6}{p}\right)\right) C_1,\\
S_4 = -S_9 &= \frac{1}{6}\left(3 - 2\left(\frac{3}{p}\right) - \left(\frac{6}{p}\right)\right) C_1,\\
S_5 = -S_8 &= \frac{1}{4}\left(-1 - \left(\frac{2}{p}\right) + \left(\frac{3}{p}\right) + \left(\frac{6}{p}\right)\right) C_1 + \frac{1}{4}C_3,\\
S_6 = -S_7 &= \frac{1}{12}\left(3 - 3\left(\frac{2}{p}\right) + \left(\frac{3}{p}\right) - \left(\frac{6}{p}\right)\right) C_1 - \frac{1}{4}C_3.
\end{aligned}
$$

For prime p, the first system gives the formulas printed in the text; in the remaining cases analogous formulas are obtained from the second system.

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